

Zhe Wang

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PROFESSIONAL SUMMARY

Business Analytics graduate student with hands-on experience building Python/SQL data pipelines, REST APIs, relational data systems, and reproducible data-processing workflows. Skilled in incremental ingestion, data quality validation, idempotent loading, automated testing, and developing reliable datasets and services for analytical and machine learning applications.

EDUCATION

University of Dayton Dayton, OH <i>M.S. in Business Analytics</i>	Aug 2024 - Expected Dec 2026 GPA: 3.65/4.0
Harbin University of Commerce Harbin, China <i>Bachelor's Degree in Exhibition Economy and Management, Minor: Finance</i>	Sep 2020 - July 2024

CERTIFICATIONS

AWS Certified Data Engineer - Associate Amazon Web Services (AWS)	Issued Aug 2026
AWS Certified Machine Learning Engineer - Associate Amazon Web Services (AWS)	Issued Jul 2026

EXPERIENCE

Data Analyst Intern <i>Richat Partners</i>	Aug 2025 - Oct 2025 Remote
- Built automated Python data pipelines integrating 10+ external APIs (IMF, World Bank), enabling reliable multi-source ingestion for downstream analytics and modeling - Designed FastAPI-based data services and reusable retrieval workflows to improve programmatic data access and reduce manual workflow effort - Standardized and validated multi-source datasets to improve data quality, consistency, and reliability across analytical workflows - Developed KPI dashboards across 4+ domains to validate downstream usability and communicate business performance to stakeholders	

PROJECTS

Public Safety and Weather Data Integration Pipeline <i>University of Dayton, Graduate Project</i>	Fall 2025
- Built a Python ETL pipeline integrating 16 months of public safety and weather data from 2 REST APIs into MySQL on Amazon RDS through historical backfill and scheduled daily ingestion - Automated incremental processing with EventBridge-triggered Lambda, database-derived MAX(date) checkpoints, and 2,000-record API pagination to avoid unnecessary reprocessing - Implemented idempotent batch writes using INSERT IGNORE and ON DUPLICATE KEY UPDATE, with Secrets Manager-managed credentials and SQL validation checks to preserve data integrity - Developed 8 analytical SQL queries, including cross-source joins, to support efficient retrieval, anomaly review, and downstream analysis	
House Prices Data & ML Pipeline <i>Independent Project</i>	Spring 2026
- Built a modular Python pipeline for 79 predictors, automating preprocessing, feature engineering, training, and evaluation across 4 base models and 2 ensembles - Created reusable scikit-learn transformers for missing values, log transforms, and categorical encoding within fold-safe data-processing workflows - Added 8 pytest unit tests and GitHub Actions CI across Python 3.10/3.11 to validate transformations, pipeline behavior, and reproducibility - Implemented versioned artifacts, metrics, data fingerprints, and deployment aliases to support repeatable training and inference workflows	
GD-OTS Corporate KPI Data Pipeline & Dashboard <i>University of Dayton, Graduate Capstone Project</i>	Spring 2026
- Transformed 27 KPI columns across 24 manufacturing sites from non-standard Excel layouts into a long-form fact table at Site x Year x Month x Metric grain using Python/pandas - Built config-driven Python scripts to standardize labels, expand annual targets to monthly grain, and flag missing metrics or duplicate sites - Modeled DimSite, DimMonth, and monthly targets in Power BI using a dimensional schema supporting a 4-level organizational hierarchy - Delivered reusable analytical views for trends, outliers, and site comparisons across Safety, Productivity, Quality, Delivery, and Cost	

SKILLS

- Programming & Databases:** Python (pandas, NumPy, FastAPI, pytest), SQL, MySQL, PostgreSQL, relational modeling, Git/GitHub
- Data Engineering:** ETL pipelines, REST API ingestion, incremental processing, API pagination, idempotent loading, batch processing, data validation, data warehousing
- Cloud & Engineering Practices:** AWS Lambda, EventBridge, RDS, S3, Glue, Redshift, Secrets Manager, automated testing, GitHub Actions CI/CD
- Analytics & ML Data Workflows:** scikit-learn pipelines, feature engineering, dimensional modeling, Power BI, model evaluation, reproducible data preparation